

Day 3: Loops

Topics: while, do-while, for, nested loops

1. To Print All Integers from 1 to N

Uses a for loop to print all integers starting from 1 up to N entered by the user.

```
#include <stdio.h>

int main() {
    int n, i;
    printf("Enter N: ");
    scanf("%d", &n);
    printf("Integers from 1 to %d:\n", n);
    for (i = 1; i <= n; i++) {
        printf("%d ", i);
    }
    printf("\n");
    return 0;
}
```

2. To Evaluate Sum of All Integers from 1 to N

Uses a for loop to accumulate the sum of all integers from 1 to N.

```
#include <stdio.h>

int main() {
    int n, i, sum = 0;
    printf("Enter N: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        sum += i;
    }
    printf("Sum of integers from 1 to %d = %d\n", n, sum);
    return 0;
}
```

3. To Evaluate Factorial of Given Integer n

Uses a for loop to calculate $n! = 1 \times 2 \times 3 \times \dots \times n$.

```
#include <stdio.h>

int main() {
    int n, i;
    long long fact = 1;
    printf("Enter n: ");
    scanf("%d", &n);
    if (n < 0) {
        printf("Factorial not defined for negative numbers.\n");
    } else {
        for (i = 1; i <= n; i++) {
            fact *= i;
        }
    }
}
```

```

        printf("Factorial of %d = %lld\n", n, fact);
    }
    return 0;
}

```

4. To Print Reverse of Given Multi-Digit Number

Uses a while loop with modulo and division to extract and print digits in reverse order.

```

#include <stdio.h>

int main() {
    int n, remainder, reverse = 0;
    printf("Enter a number: ");
    scanf("%d", &n);
    int original = n;
    while (n != 0) {
        remainder = n % 10;
        reverse = reverse * 10 + remainder;
        n /= 10;
    }
    printf("Reverse of %d = %d\n", original, reverse);
    return 0;
}

```

5. To Count Digits in a Given Number

Uses a while loop to repeatedly divide the number by 10 and count iterations.

```

#include <stdio.h>

int main() {
    int n, count = 0;
    printf("Enter a number: ");
    scanf("%d", &n);
    if (n == 0) {
        count = 1;
    } else {
        while (n != 0) {
            n /= 10;
            count++;
        }
    }
    printf("Number of digits = %d\n", count);
    return 0;
}

```

6. To Check Whether Entered Number is Prime or Not

Uses a for loop to check if any number from 2 to $\sqrt{n}$ divides n evenly.

```

#include <stdio.h>
#include <math.h>

int main() {
    int n, i, isPrime = 1;
    printf("Enter a number: ");
    scanf("%d", &n);

```

```
if (n <= 1) {
    isPrime = 0;
} else {
    for (i = 2; i <= (int)sqrt(n); i++) {
        if (n % i == 0) {
            isPrime = 0;
            break;
        }
    }
}
if (isPrime)
    printf("%d is a Prime Number\n", n);
else
    printf("%d is NOT a Prime Number\n", n);
return 0;
}
```

— End of Day 3 Assignment —